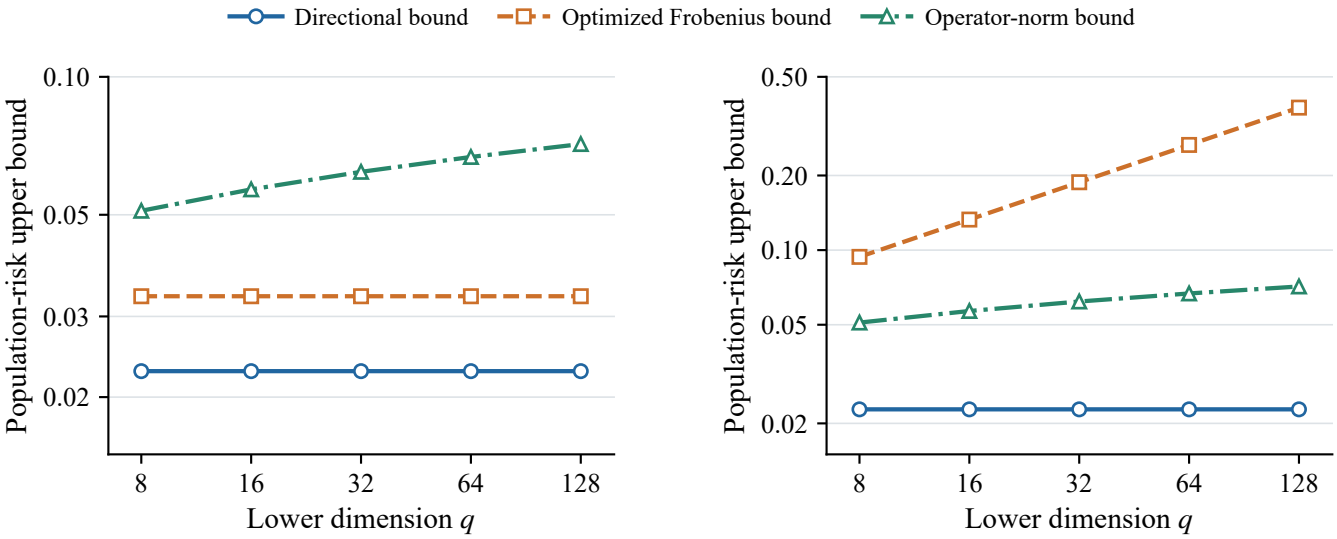
Directional bound
Optimized Frobenius bound
Operator-norm bound
Population-risk upper bound
0.10
0.05
0.03
0.02
8
16
32
64
128
Lower dimension q
(a) Diagonal
Population-risk upper bound
0.50
0.20
0.10
0.05
0.02
8
16
32
64
128
Lower dimension q
(b) Two-basis